

Multiple Choice

1. **Answer:** D

Options: A) a cell wall made of cellulose; B) lysosomes; C) large vacuoles that store water; D) centrioles within centrosomes

Note: Correct option: D - centrioles within centrosomes

2. **Answer:** E

Options: A) auxins; B) neurotransmitters; C) enzymes; D) hormones; E) pheromones

Note: Correct option: E - pheromones

3. **Answer:** A

Options: A) Carbon dioxide bubbles forming in the blood causing acidosis; B) Rapid increase in barometric pressure causing the bends; C) Formation of helium bubbles in the bloodstream due to rapid ascent; D) Increased salinity in the water leading to dehydration and joint pain; E) Oxygen bubbles forming in the blood leading to hypoxia

Note: Correct option: A - Carbon dioxide bubbles forming in the blood causing acidosis

4. **Answer:** A

Options: A) homeotic genes; B) cell division; C) signal transduction; D) cell differentiation; E) apoptosis

Note: Correct option: A - homeotic genes

5. **Answer:** A

Options: A) multiple copies of a single chromosome; B) not visible even under a light microscope; C) produced only during mitosis; D) produced by repeated rounds of nuclear division without DNA replication; E) single copies of multiple chromosomes

Note: Correct option: A - multiple copies of a single chromosome

True / False

6. **Answer:** False

Options: A) True; B) False

Note: While more than a quarter of the middle-aged adult population said they were interested in PAS insurance, actual participation would be highly dependent on premium rates. The current lack of publicly subsidized insurance for long-term care and personal assistance services remains a serious gap in the disability service system.

7. **Answer:** True

Options: A) True; B) False

Note: The gluten-free diet has minimal deficiencies, similar to those present in the diet with gluten, with an improvement in the lipid profile by increasing the proportion of monounsaturated fatty acids to the detriment of saturated fatty acids.

8. **Answer:** True

Options: A) True; B) False

Note: Ketamine sedation was successful and well tolerated in all cases. The use of atropine as an adjunct for intramuscular ketamine sedation in children significantly reduces hypersalivation and may lower the incidence of post-procedural vomiting. Atropine is associated with a higher incidence of a transient rash. No serious adverse events were noted.

9. **Answer:** False

Options: A) True; B) False

Note: While FS was commonly used in patients undergoing thyroidectomy at our institution, in no patient over the last decade did FS correctly alter the intraoperative management. Given the time required to perform FS and the cost associated with it, we believe that routine FS should not be performed in these patients.

10. **Answer:** True

Options: A) True; B) False

Note: It is important to ensure that new mothers are adequately informed about topics important to them while in hospital. The findings highlight the need for accessible and appropriate community-based information resources for women in the postpartum period, especially for those of low socioeconomic status.

Long Form Response

11. **Answer:** Gymnosperm means 'naked seeds' and angiosperm means 'enclosed seeds'. Conifers are distinguished from flowering plants by their ability to grow in colder climates and their distinct cone-like fruits.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** M phase is the phase for cell growth, G₀ is a phase for temporary cell arrest, G₁ is for organelle duplication, S phase is for RNA transcription, and G₂ is the phase before cell division.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.